Session Objectives

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1. Identify the lobes, nuclei, and peduncles of the cerebellum along with associated brainstem nuclei.
2. Identify the pathways linking the cerebellar cortex and nuclei to the spinal cord, brainstem, and cerebrum.
3. Describe the basic cerebellar circuit, the input, internal organization, and output.
4. Understand how the functional specialization of different cerebellar regions relate to various abilities along with the origin and expression of cerebellar disorders.

Nolte's The Human Brain: An introduction to its functional anatomy

Chapter 20 Cerebellum

http://arktos.nyit.edu/login?url=https://www-clinicalkey-com.arktos.nyit.edu/#!/content/book/3-s2.0-B9780323653985000205

General features of the cerebellum

- “Little brain” composed of cortical sheet and nuclei
  - 80% of surface area of cerebral cortex!
  - 80% of neurons in the brain!
- Critical for motor coordination.
- Damage results in poor coordination.
- Does not directly drive motor neurons.
- Less clear relation to cognitive & affective function.

Gordon Holmes studied wounded British soldiers (WW1); the standard helmet did not protect the back of their heads unlike German solders.

Holmes GM (1917) The symptoms of acute cerebellar injuries due to gunshot injuries. Brain 40:461–535

Impressively understudied

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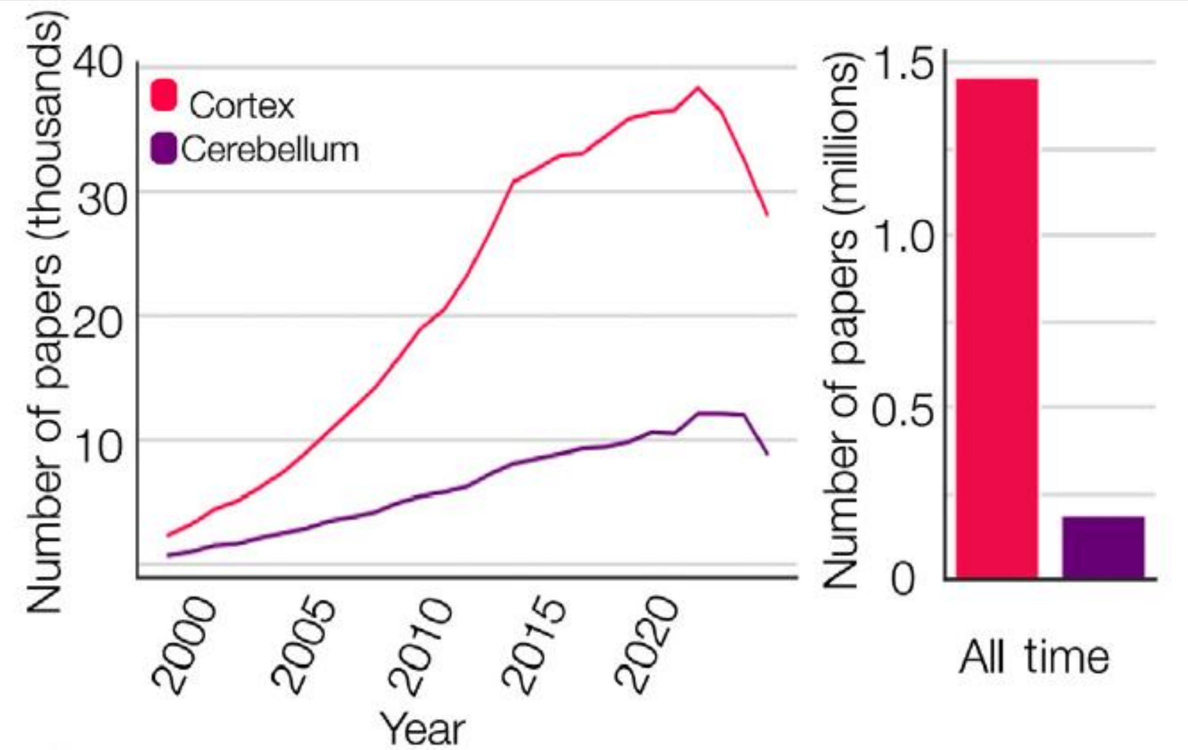
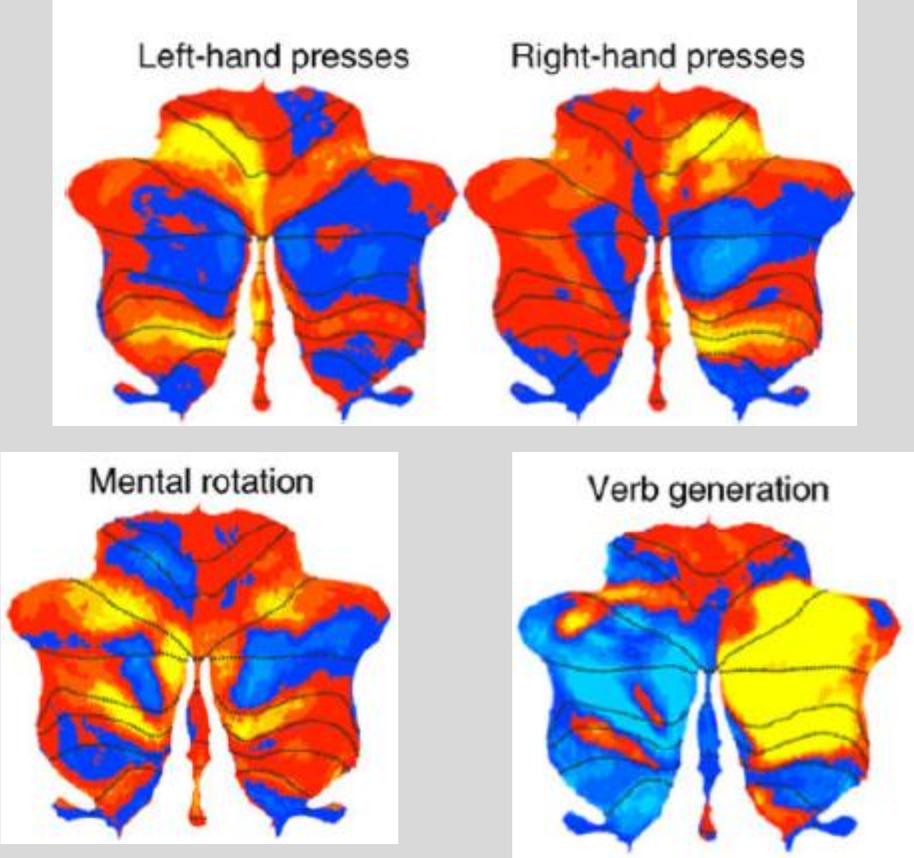
- about 10% of the research on cerebral cortex
- very few studies examine the variety of tasks needed to uncover cognition and emotion



Wang (2025) Ignoring the cerebellum is hindering progress in neuroscience



King (2019) Functional boundaries in the cerebellum revealed by multi-domain task battery

Gross cerebellar anatomy

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- Lobes and regions - functional divisions of the cortical sheet
- Peduncles - fiber tracts conveying info to and from
- Internal Nuclei – the primary output nuclei of the cerebellum
- Associated external nuclei – input nuclei to the cerebellum

The internal nuclei (deep cerebellar nuclei)

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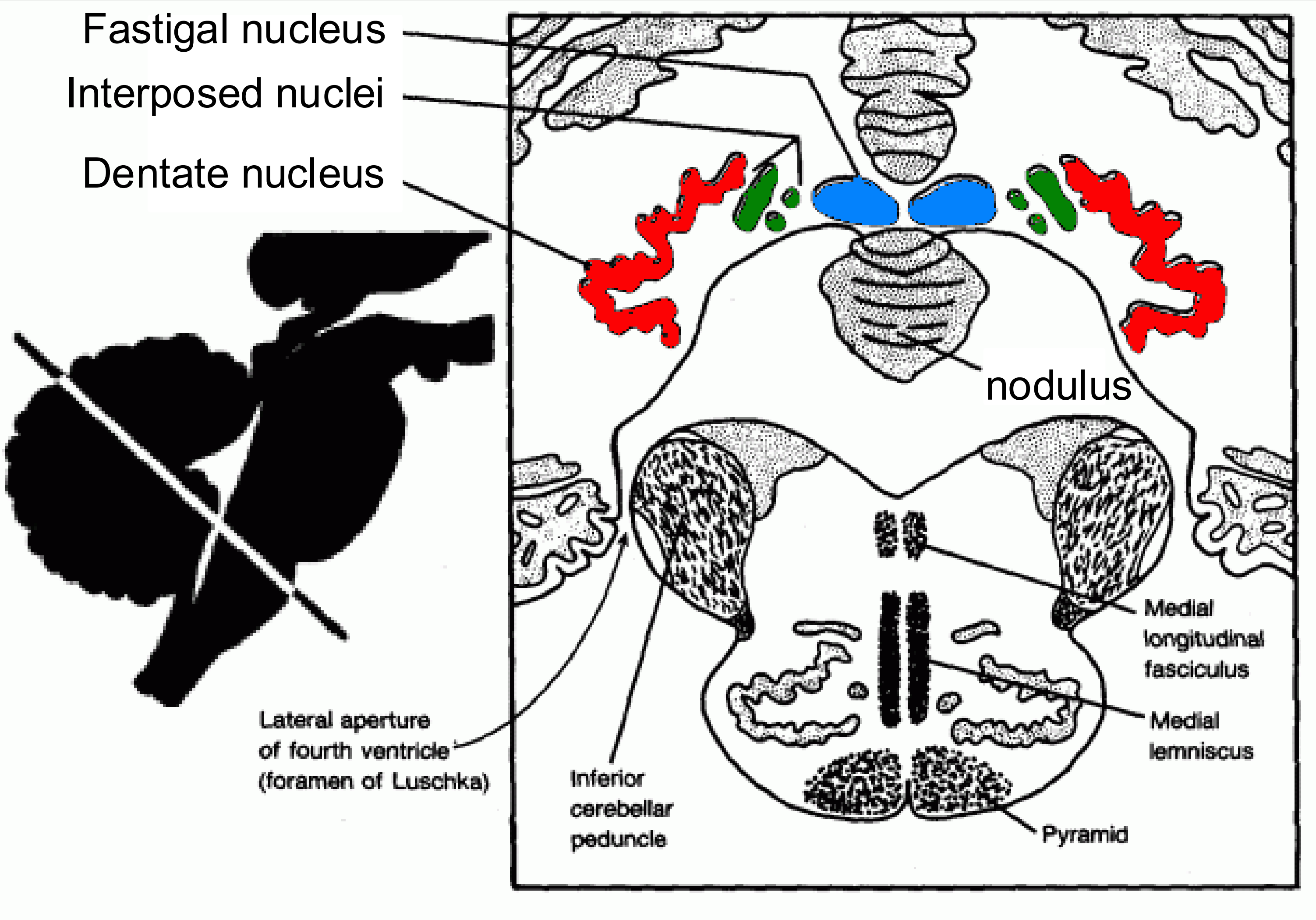

Relay information <u>from</u> the cerebellar cortex <u>to</u> brainstem and cerebral cortex

The internal nuclei (deep cerebellar nuclei)

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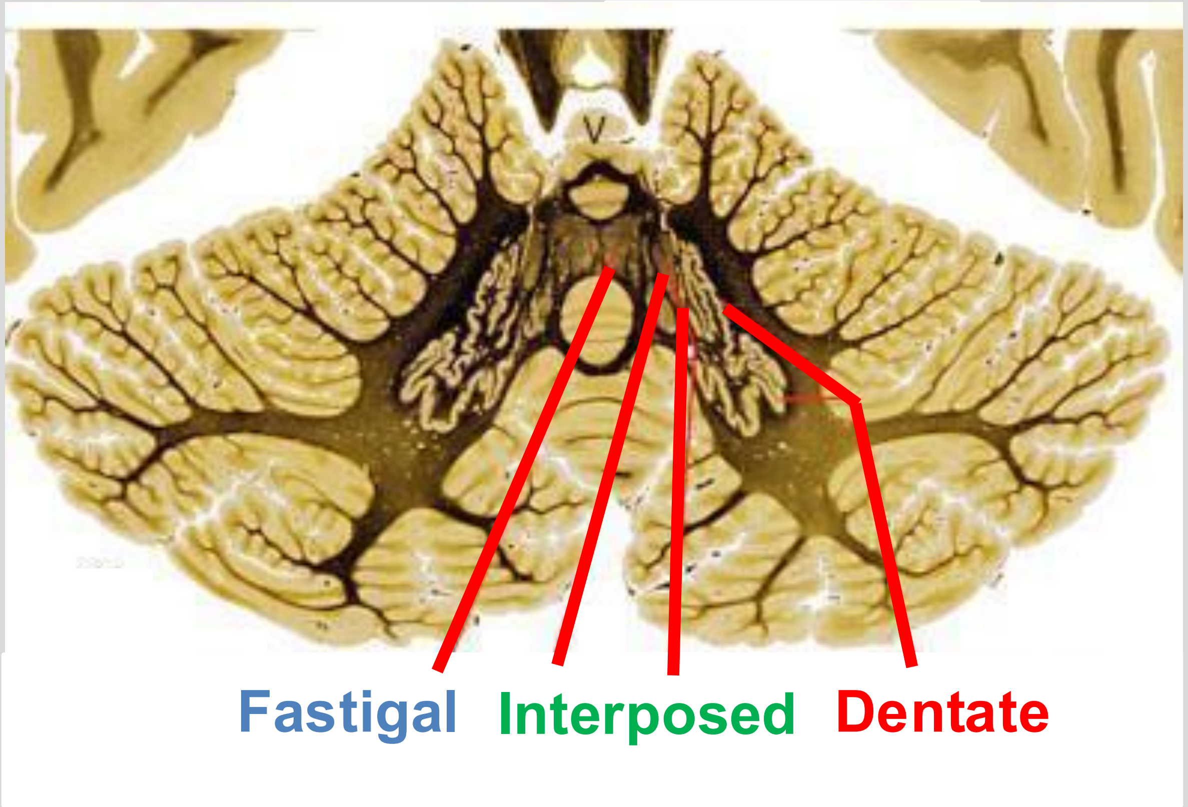

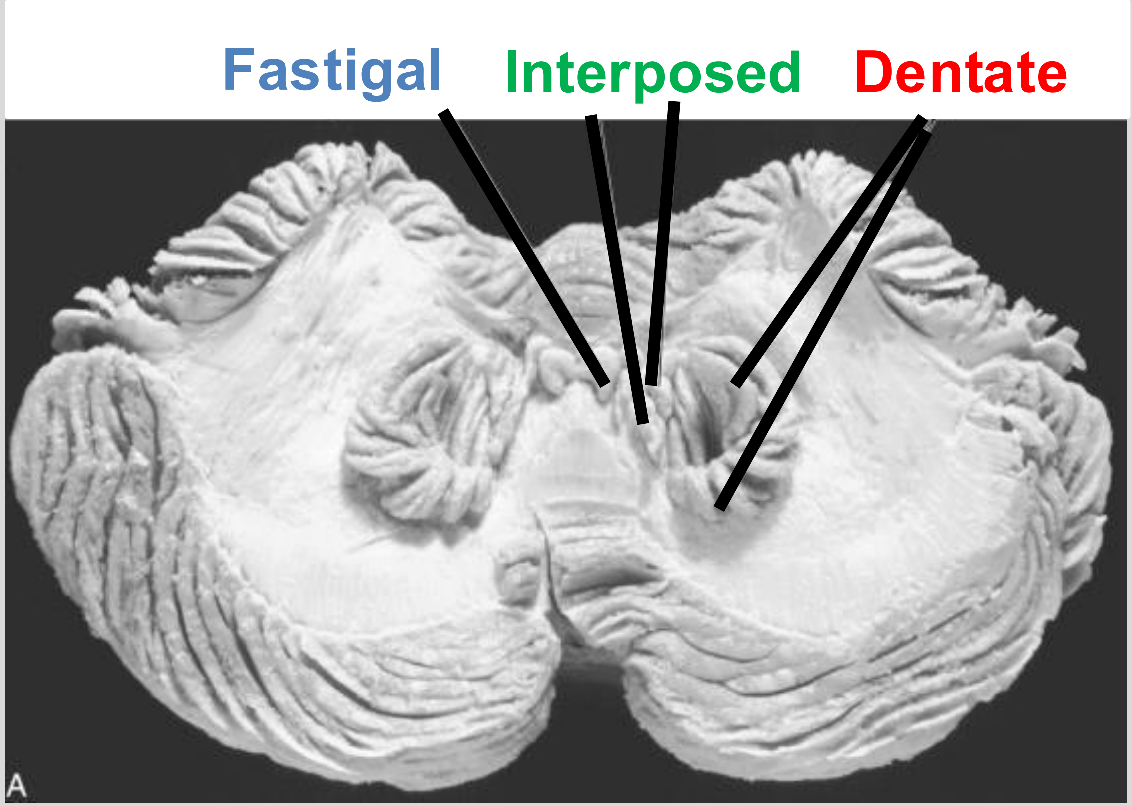

All output (with one exception) are axons from the deep cerebellar nuclei.

Relay information to the brainstem and cerebral cortex.

External brainstem nuclei linked to cerebellum

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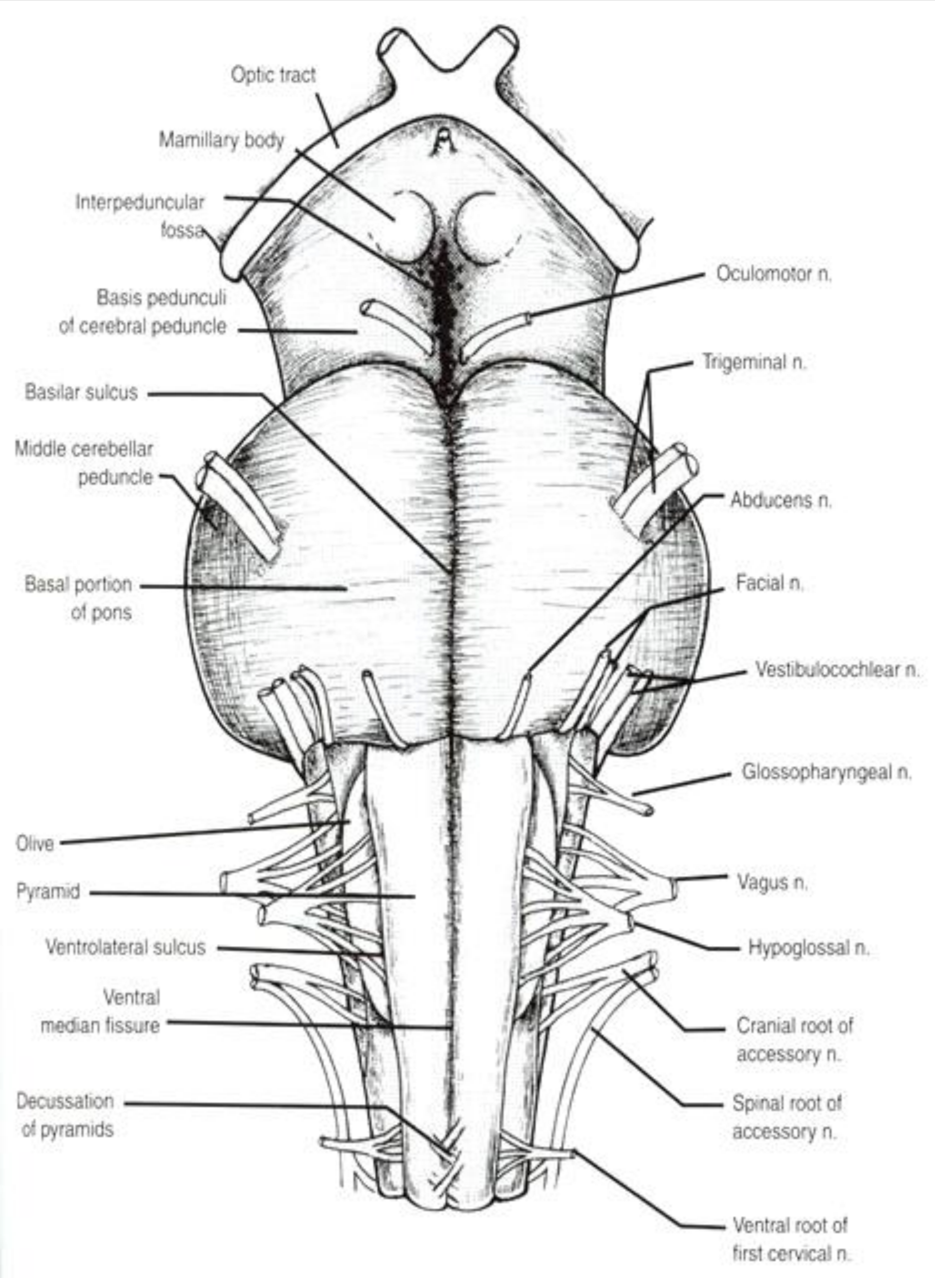

Red nucleus (midbrain) – receives output from the contralateral cerebellum

Pontine nuclei (pons) – transmits information from ipsilateral cerebrum to contralateral cerebellum

Vestibular nucleus (medulla) – receives output from ipsilateral cerebellum

Inferior Olive (medulla) - send inputs/afferents to contralateral cerebellum

External brainstem nuclei linked to cerebellum

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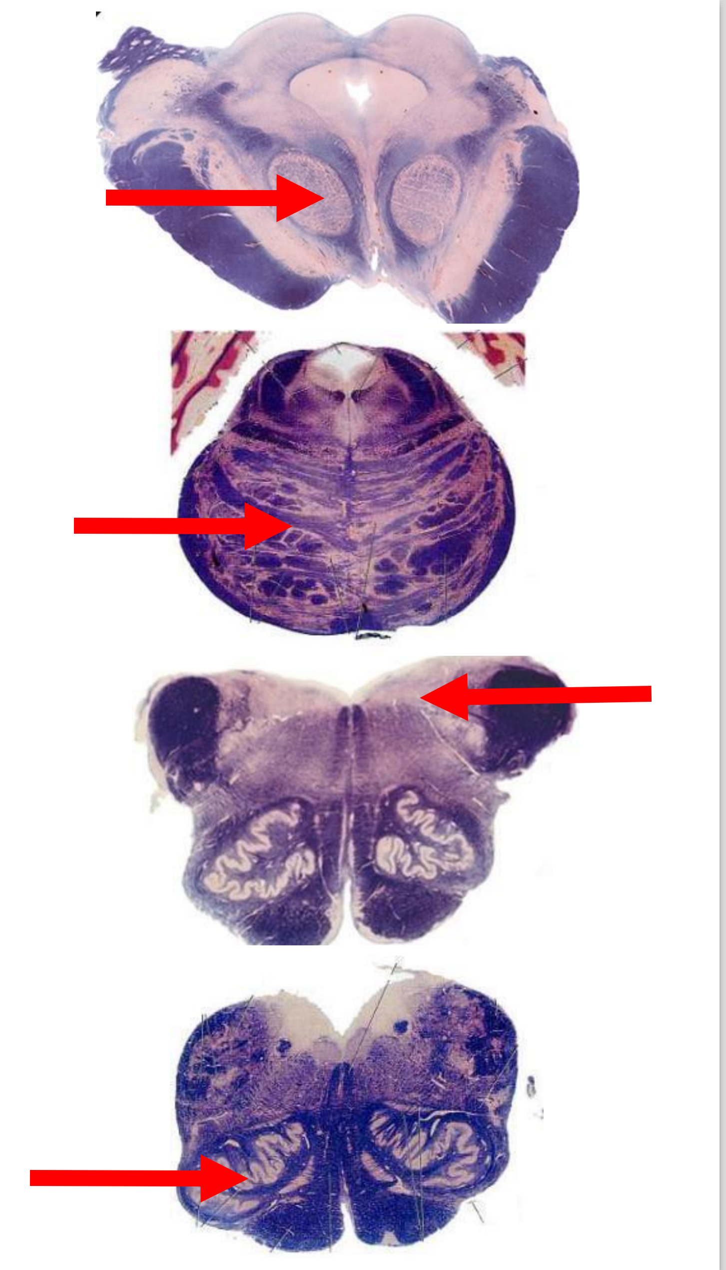

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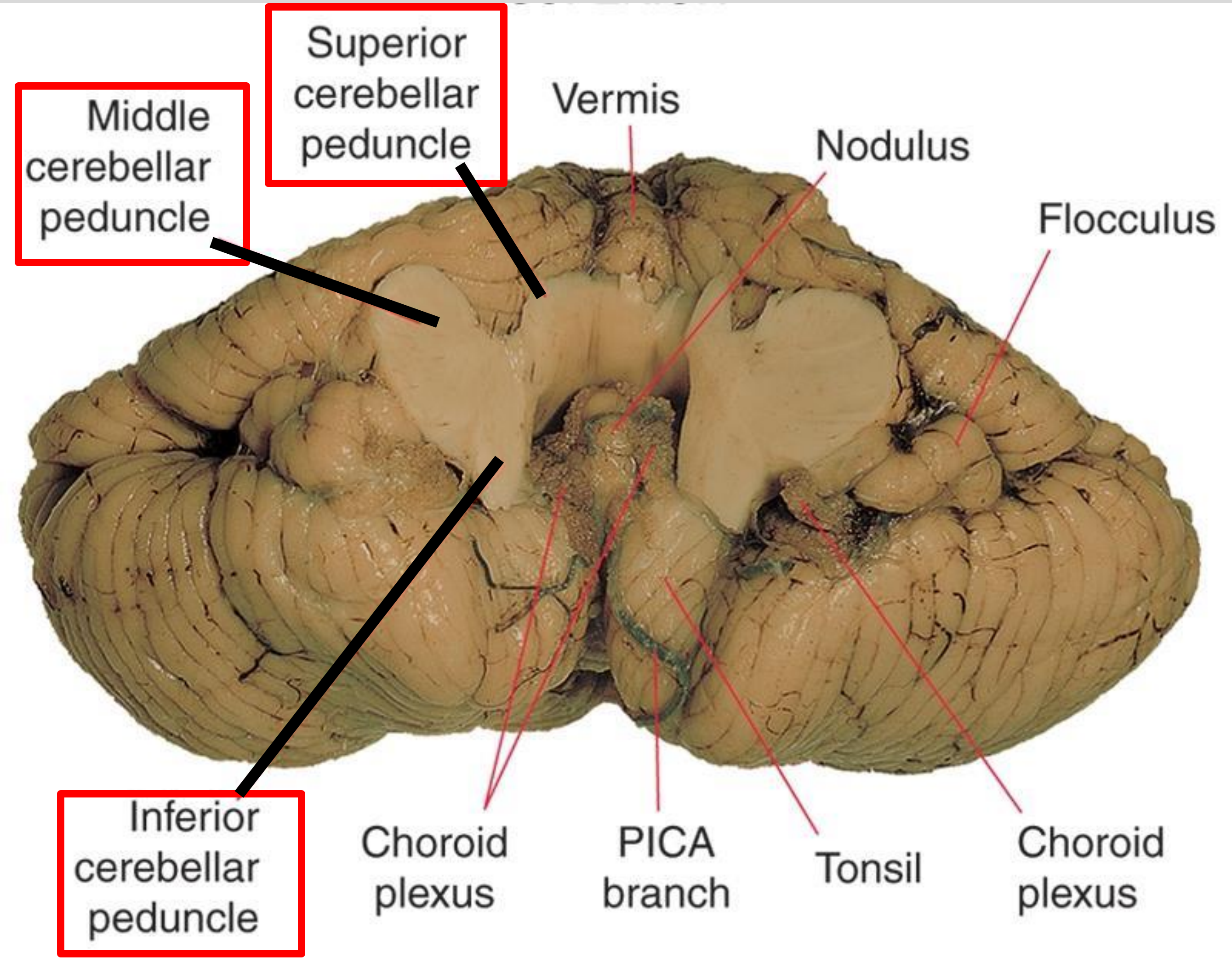
Cerebellar peduncles



(A from Nolte J: The human brain, ed 5, St. Louis, 2002, Mosby.)

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Cerebellum communicates with three major portions of the nervous system

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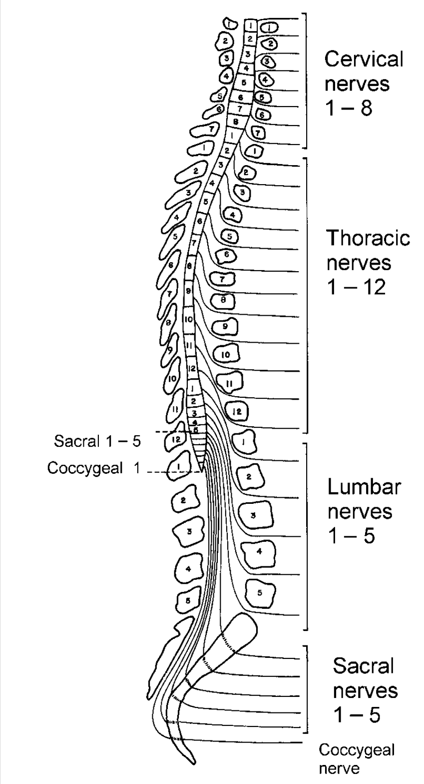
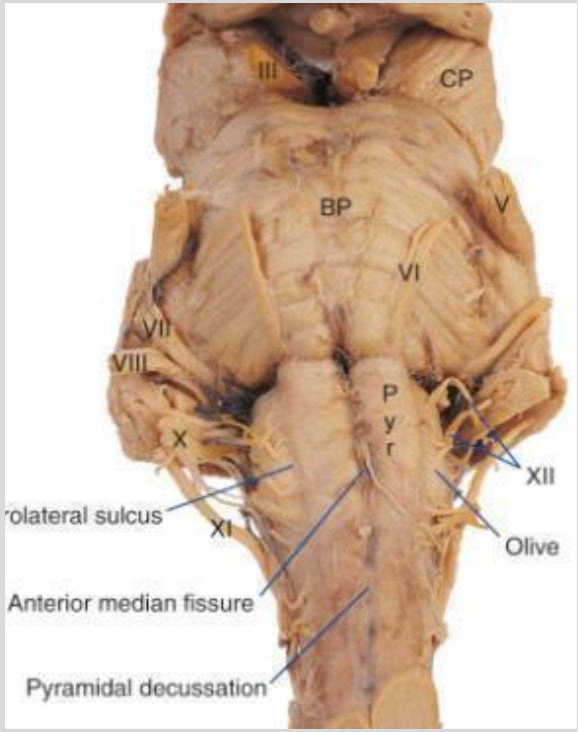
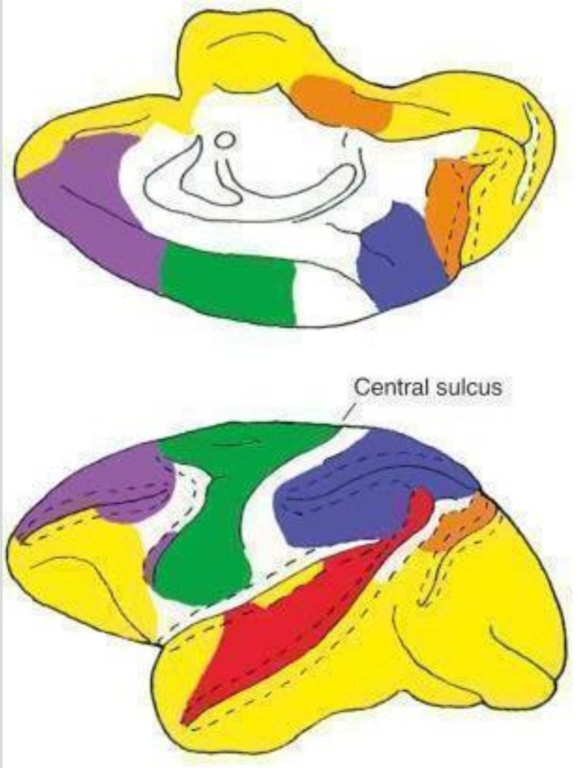
Spinal Cord



Brainstem



Cerebral cortex



Remember that cerebellum controls the ipsilateral body. Hence, signals (to and from) must cross side to communicate with the cerebellum. A mix from the spinal cord and brainstem.

Three functional divisions of the cerebellum

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Vestibulocerebellum

Flocculonodular lobe receives input from vestibular nuclei and nerves.

Head-eye coordination and balance.

Spinocerebellum

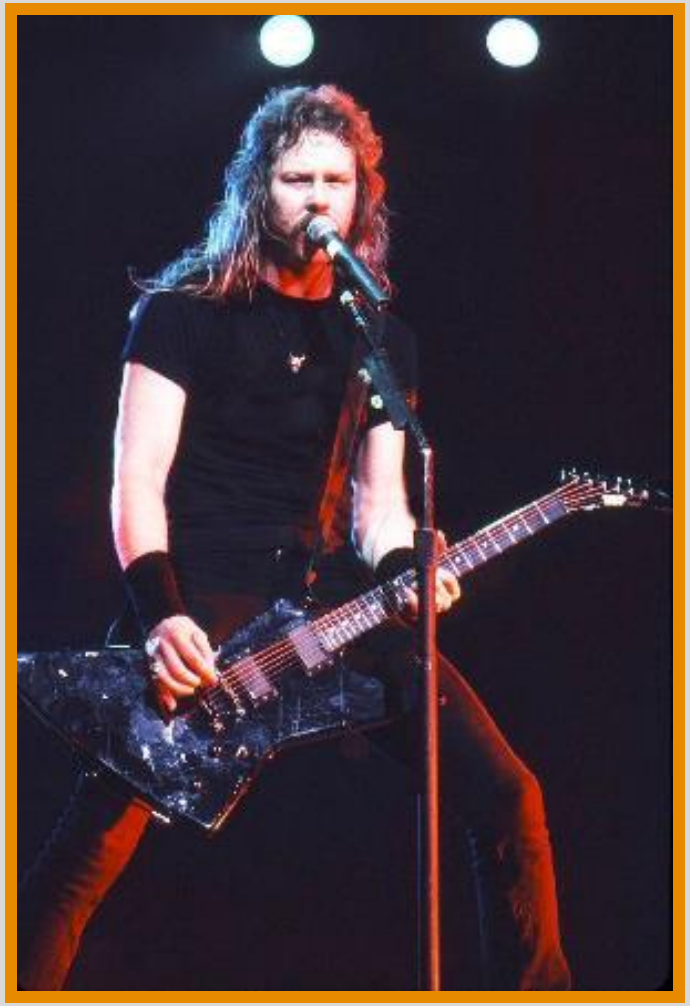
Vermal and paravermal regions of anterior & posterior lobes receive spinal inputs.

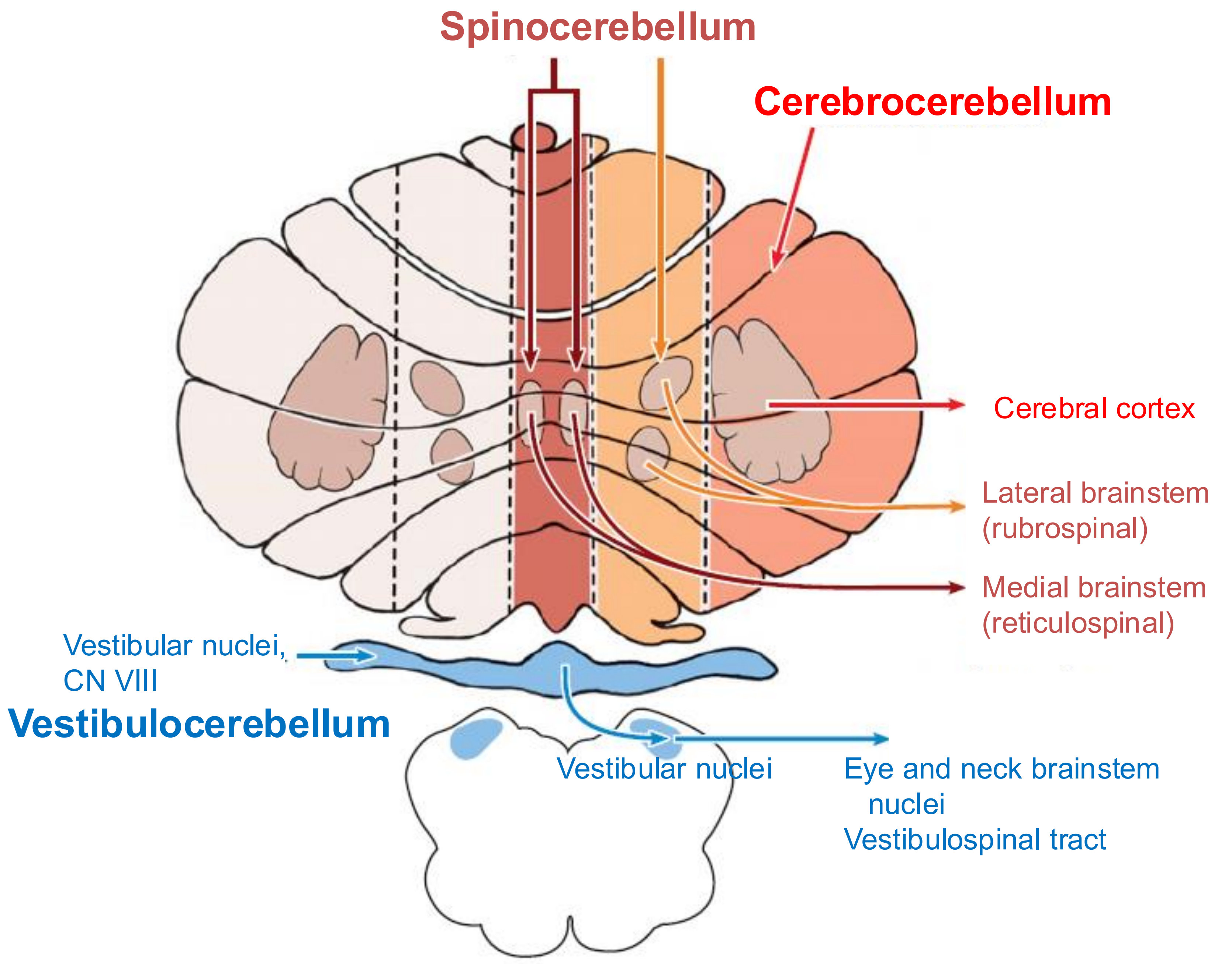
Axial control and gait.

Cerebrocerebellum

Lateral hemispheres of anterior and posterior lobes receive cerebral cortical input via pontine nuclei.

Voluntary movements especially learned and skillful movements





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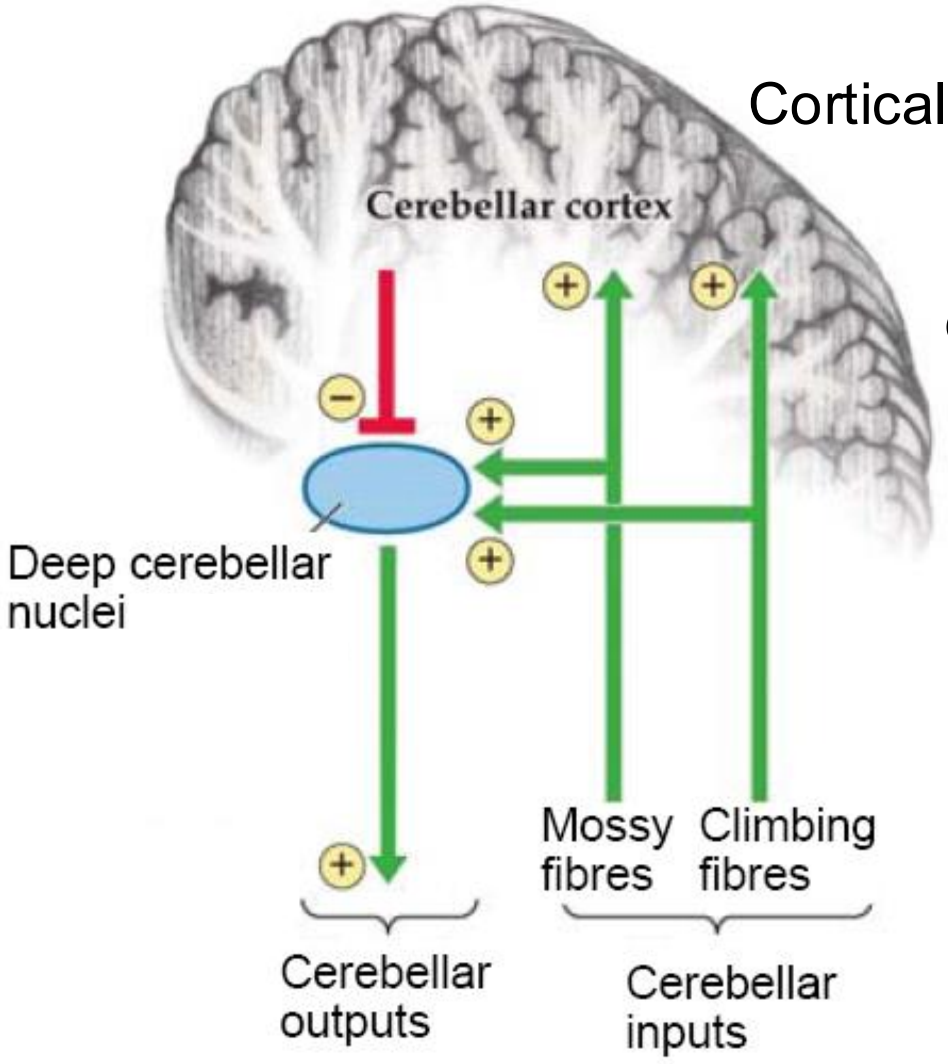
Functional divisions of the cerebellum



Cortical Processing

Purkinje cells project to deep cerebellar nuclei; inhibitory (GABA).

Mossy Fibers – afferents from brainstem nuclei, spinal cord and cerebrum; excitatory (glutamate).

Climbing fibers – afferents of the olivocerebellar tract; excitatory (glutamate).

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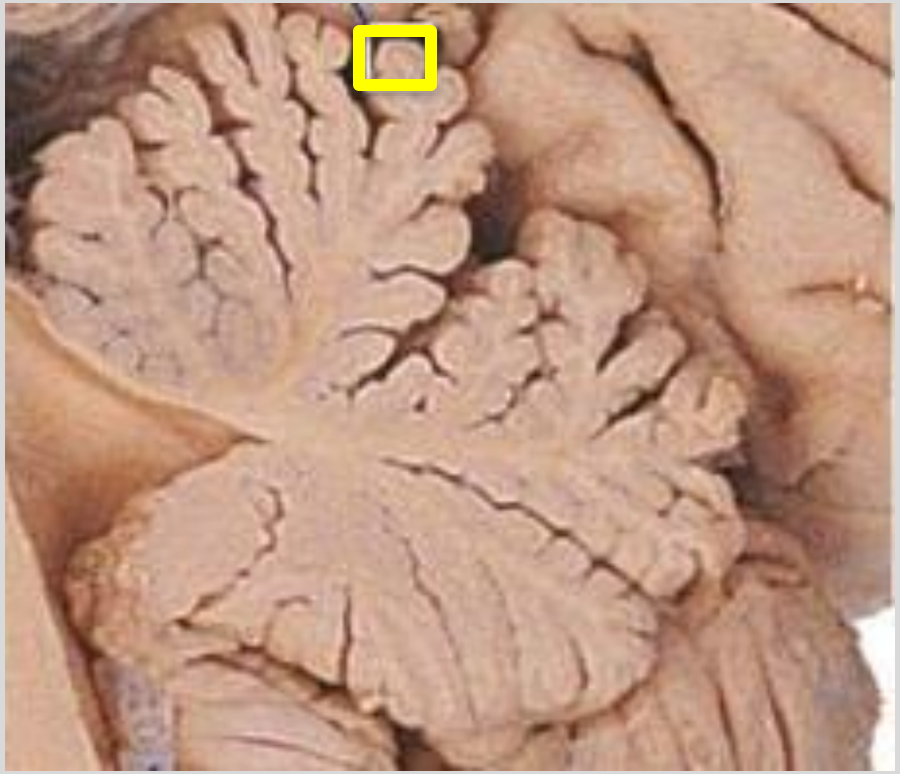
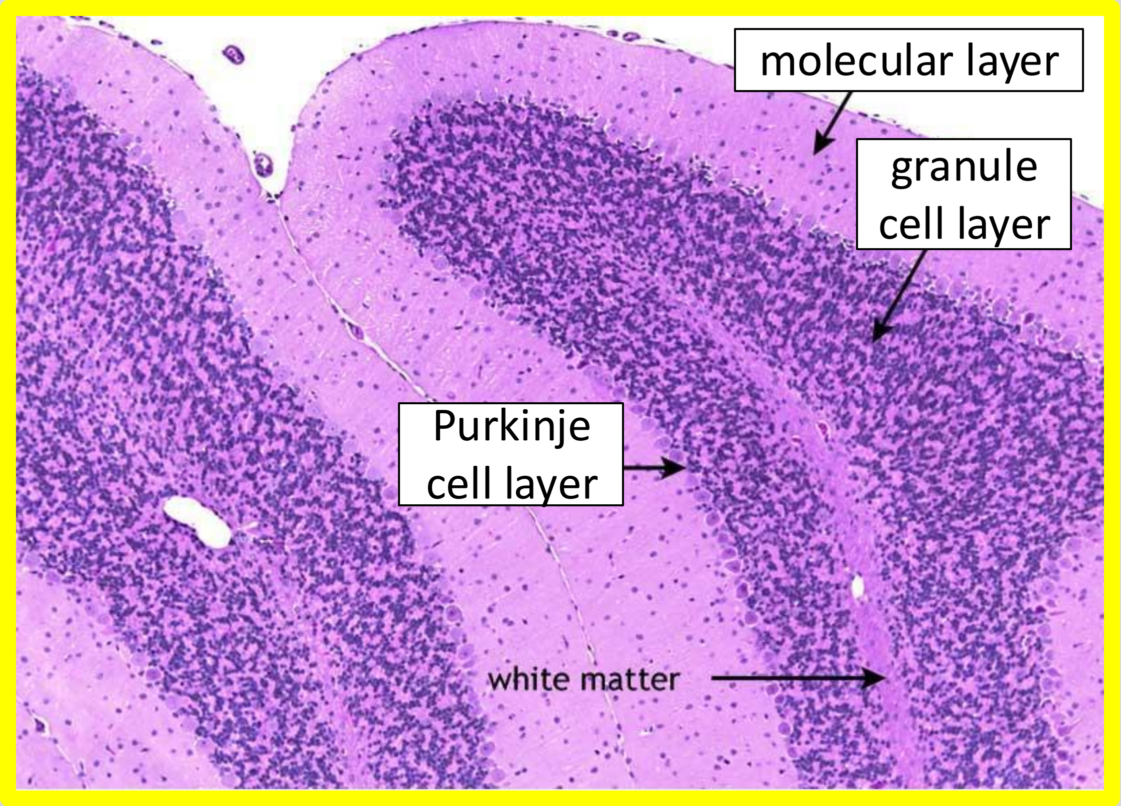
Basic cerebellar circuit

Deep cerebellar nuclei receives excitatory input (raw/"measured state" of the body and motor commands) and a processed inhibitory input (processed/"predicted state" of the body) and the two are compared.

Layers of cerebellar cortex





Molecular layer is closest to the surface - contains dendrites of Purkinje cells and axons of granule cells

Granule layer is the deepest layer and <u>more neurons than all others in the nervous system</u>.

Purkinje cell layer is between the two; <u>one layer of Purkinje cell bodies</u>.
Exclusive output from the cerebellar cortex. GABAergic/inhibitory.

Summary of internal connections

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- Climbing fibers
  - olivocerebellar tract
  - excite cerebellar cortex and deep cerebellar nuclei
- Mossy fibers
  - all other tracts
  - excite cerebellar cortex and deep cerebellar nuclei
- Purkinje cells
  - output of cerebellar cortex
- Deep cerebellar nuclei
  - excited by mossy fibers and climbing fibers ("raw" measured state)
  - inhibited by Purkinje cells (highly processed "predicted state")
  - difference of this two is the final output of cerebellum

Cerebellar disorders can result from a wide variety of causes

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Environmental: alcohol abuse, lithium, gluten intolerance

Acute diseases: tumors, stroke (<u>lateral medullary syndrome</u>), trauma, & viral infections (chicken pox)

Congenital malformations: Arnold-Chiari malformations & Dandy-Walker
Progressive hereditary disorders: Friedreich's ataxia, multiple sclerosis

ATAXIA is the most common symptom - dyscoordination in timing & magnitude -> overshoots, misdirected movements, and oscillations.

Not weakness (palsy, paralysis), or spasticity!

Similar to alcohol intoxication and are scrutinized in "Standard Field Sobriety Test".

Damage to just one region is rare, e.g. each cerebellar artery supplies different functional divisions.

<u>Damage specifically on one side would impact coordination on the same side.</u>

<u>Cerebellum controls the ipsilateral body.</u>

Cerebrocerebellar Dysfunction

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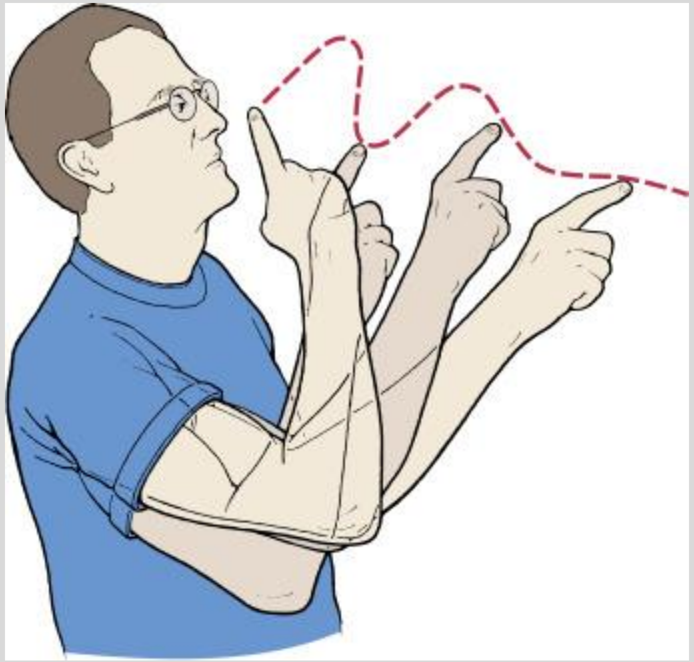
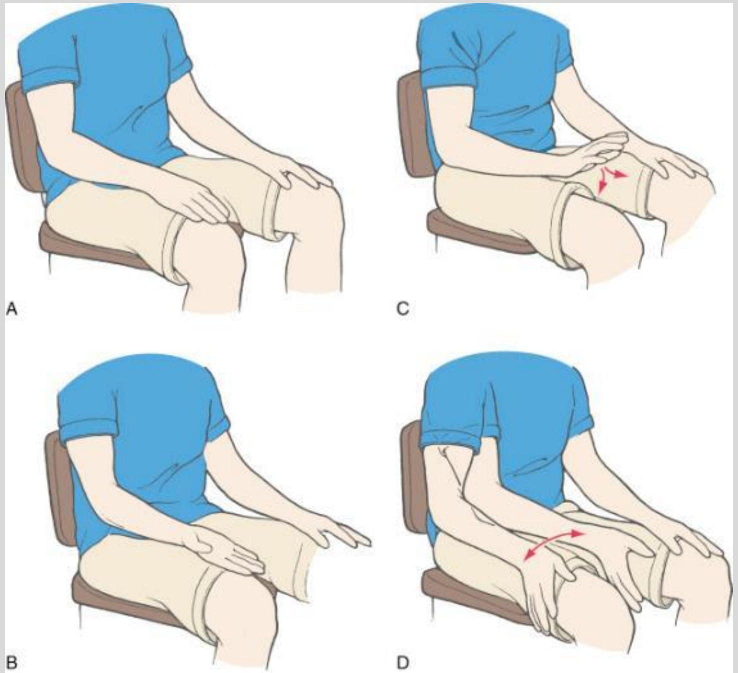
Intention tremor



Problems with rapid alternating movements*



Symptoms – Ataxia incl. misdirections and overshoots by the upper limb
– Intention Tremor (increasing tremor as hand approaches the movement target) NOT resting tremor
– *Dysdiadochokinesia (DDK), pronunciation → dis-die-a-dah-ko-kinesia
– Slurred speech
– Hypotonia (low muscle tone)

Causes – Head trauma affecting dentate nuclei or lateral hemisphere, pontine degeneration

Summary

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- Cerebellum is composed of a cortical sheet and internal nuclei.
- Processes information from the spinal cord, brainstem and cerebral cortex.
- Functional divisions defined by cortical territory, output nuclei, and capabilities.
- Information processing involves mossy fibers, climbing fibers, granule cells, Purkinje cells and deep cerebellar nuclei.
- Cerebellum is essential for coordination. Dysfunction is manifest as ataxia.